

# Multiparticle Dynamics

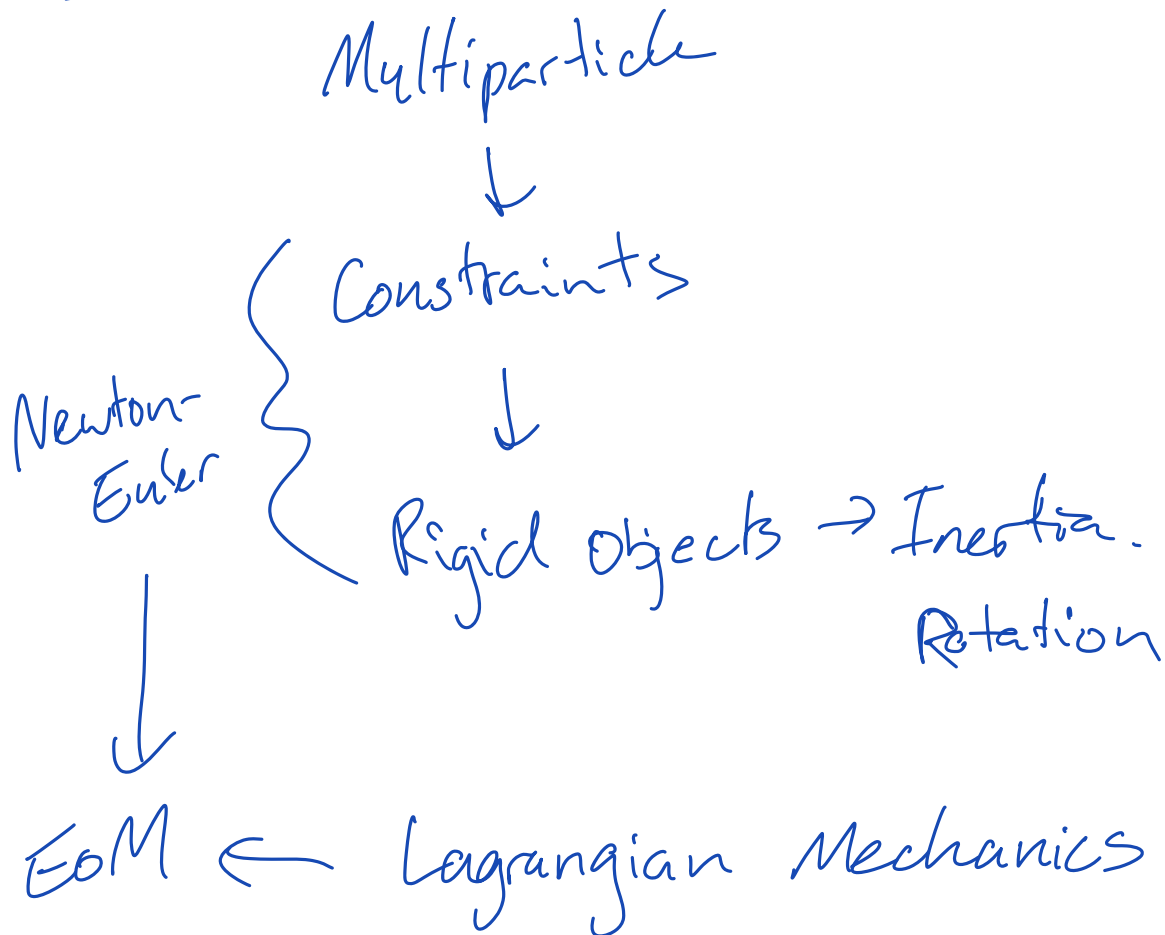
Prelim 1.  
10/4-8

Prelim 2  
Week 12.

- Overview
- 2-Body Problem
- 3-Body Problem
- N-Body Problem

HW4 due  
Friday 6pm.  
Q3. a) - e)  
b) draw FBD.

## Overview

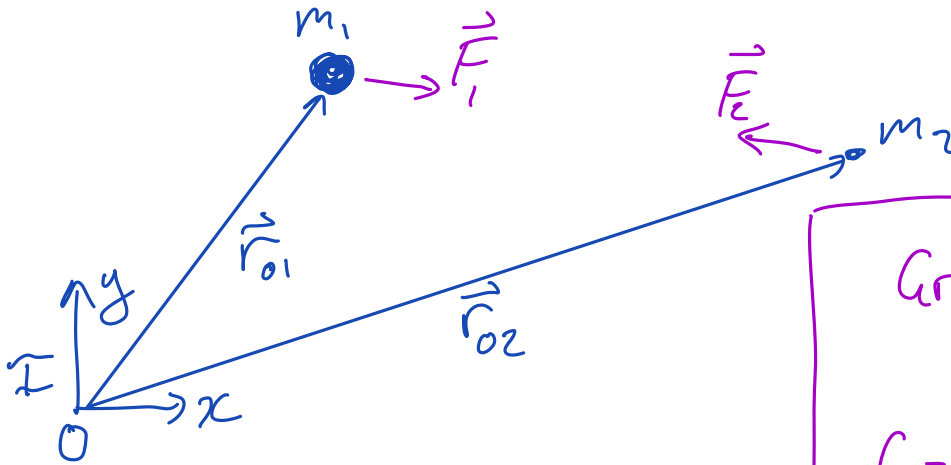


Prelim 2

Complex → Final project  
→ Exam.

# The 2-Body Problem

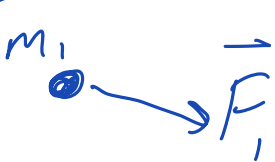
$$\vec{F}_1 = -\vec{F}_2$$



Gravity  $F_{12} = \frac{Gm_1m_2}{|r_{12}|^2}$

$$G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

FBD



$$\vec{F}_1 = \frac{Gm_1m_2}{|r_{12}|^2} \cdot \frac{\vec{r}_{12}}{|r_{12}|} = \frac{Gm_1m_2 \vec{r}_{12}}{|r_{12}|^3}$$

$$= \frac{Gm_1m_2 \vec{r}_{12}}{|r_{12}|^3}$$



$$\vec{F}_2 = \frac{Gm_1m_2}{|r_{21}|^2} \cdot \frac{\vec{r}_{21}}{|r_{21}|} = \frac{Gm_1m_2 \vec{r}_{21}}{|r_{12}|^3}$$

$$= \frac{Gm_1m_2 \vec{r}_{21}}{|r_{12}|^3}$$

norm() →

EOM

$$\dot{\vec{r}}_1 = \vec{v}_1$$

$$\dot{\vec{r}}_2 = \vec{v}_2$$

$$\dot{\vec{v}}_1 = \vec{F}_1/m_1$$

$$\dot{\vec{v}}_2 = \vec{F}_2/m_2$$

Matlab

$$\Rightarrow \dot{x} = f(x) = f \begin{pmatrix} x_1 \\ y_1 \\ x_2 \\ y_2 \\ v_{x1} \\ v_{y1} \\ v_{x2} \\ v_{y2} \end{pmatrix}$$

# Expectations

$$\vec{H}_{/I} = \text{constant} \Rightarrow \dot{\vec{H}}_{/I} = 0$$

$$\underline{LMB} \Rightarrow \sum \vec{F}_{Ext} = \dot{\vec{P}}$$

$$0 = m_1 \vec{a}_1 + m_2 \vec{a}_2$$

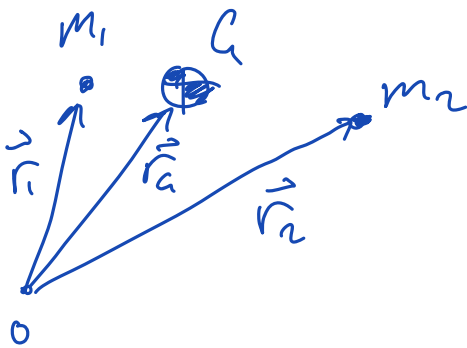
$$= \frac{d}{dt} (m_1 \vec{v}_1 + m_2 \vec{v}_2)$$

$$= \frac{d^2}{dt^2} (m_1 \vec{r}_1 + m_2 \vec{r}_2)$$

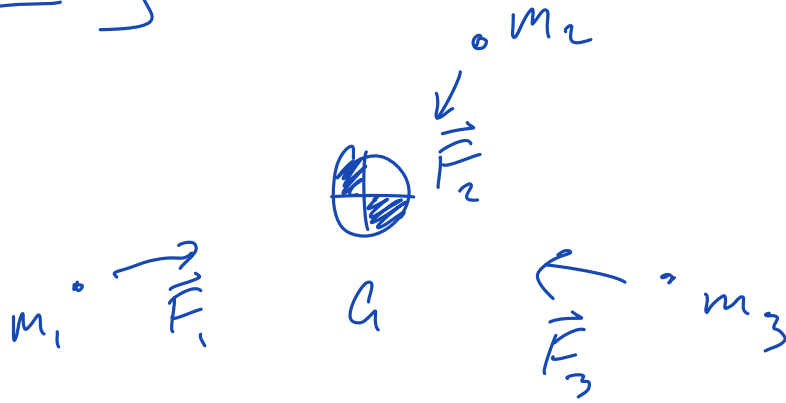
$$0 = \frac{d^2}{dt^2} (m_{TOT} \vec{r}_a)$$

$$\Rightarrow \vec{v}_a = \text{const}, \Rightarrow \vec{p}_a = \text{const}$$

We can set  $\vec{v}_a = 0$  if set  $\vec{r}_a = 0$  as our inertial frame.



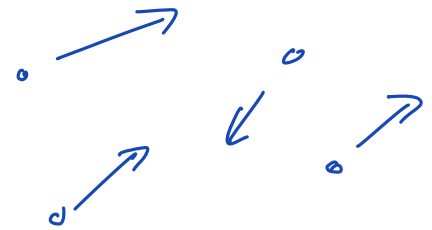
## 3-Body Problem



No analytical solution!

## N-Body Problem

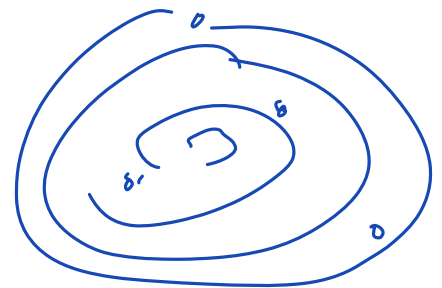
• Rarefied Gas Dynamics



• Molecular Dynamics



• Celestial Mechanics



⇒ • Rigid Bodies

